

Morphology-Adaptive Stacked Intelligent Metasurfaces for Contactless Cardiopulmonary Radar Monitoring for 6G Healthcare

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Abstract

Contactless radar can recover respiration and heartbeat from chest-wall micromotion, but the weak cardiac component is vulnerable to multipath, nearby motion and morphology-dependent scattering. We propose a morphology-adaptive stacked intelligent metasurface (SIM) that performs phase-only processing of the reflected field before a single detector. Programmable phase masks are interleaved with Fresnel coupling operators and optimized with an exact electromagnetic adjoint gradient, forming a linear wave-domain neural processor. Body mass index (BMI) is an observable conditioning variable for a parametric morphology surrogate, not a clinical law. A reproducible 30-GHz digital twin compares single-antenna, 36-channel digital-array, one-layer surface, universal three-layer SIM and BMI-adaptive three-layer SIM receivers. At 8-dB reference signal-to-noise ratio, the adaptive SIM attains 9.29-dB mean spatial signal-to-interference-plus-noise ratio, versus 9.32 dB for the digital array, 8.38 dB for the universal SIM, 5.72 dB for the one-layer surface and **−9.90** dB for a single antenna. Its respiration- and heart-rate root-mean-square errors are 0.05 breaths/min and 0.08 beats/min. Six-bit phase control approaches continuous control, while two layers maximize the tested SINR with 0.5-dB loss per layer. [These findings are limited to the specified simulation model and provide a testable system hypothesis; they do not establish clinical accuracy or full-wave hardware performance.](#)

Keywords: 6G healthcare, contactless vital-sign monitoring, Doppler radar, electromagnetic neural network, stacked intelligent metasurface, wave-domain processing

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1 Introduction

Radar is attractive for physiological monitoring because it senses motion without requiring an electrode, chest strap or wearable device. Continuous-wave (CW), frequency-modulated continuous-wave (FMCW) and ultra-wideband systems have all been used to recover breathing and cardiac activity [1–3]. Detecting a person, however, is much easier than recovering a stable heartbeat waveform. In a quiet measurement the respiratory displacement is usually dominant. Its harmonics, small posture changes, random body motion and I/Q offsets can obscure the weaker cardiac component [1, 4]. This remains a practical difficulty outside tightly controlled laboratory settings. A postoperative FMCW study, for example, obtained better respiratory agreement after movement-contaminated intervals were removed, but reported poorer accuracy during spontaneous breathing [5].

Spatial processing can help when the chest return and unwanted motion arrive from different directions. The conventional solution is a digital antenna array, which is flexible but requires a receiver and data-conversion path for each spatial channel. For a compact 6G healthcare sensor, that hardware cost may be difficult to justify. Our interest is therefore in moving part of the spatial processing in front of the receiver, where the electromagnetic field itself can be shaped before it is sampled.

A stacked intelligent metasurface (SIM) provides one possible route. Its programmable layers are separated by free-space regions, so phase modulation at one layer is mixed by diffraction before reaching the next. This cascaded operation has been studied for wave-domain beamforming, interference suppression and sensing [6–9]. The idea is closely related to diffractive neural processors demonstrated at optical and microwave frequencies [10, 11]. We use the neural-network analogy only in this limited sense: the phase states are trainable and the intervening propagation couples the elements. A passive narrowband stack is still a linear complex-field operator; it does not contain a physical nonlinear activation.

An additional issue is that the desired field need not be identical for every person. Measurement position and subject anatomy affect radar signal quality, as is also evident in public synchronized datasets [12, 13]. We use body mass index (BMI) as a readily available conditioning variable, while recognizing that BMI is only a coarse proxy for body morphology. It cannot determine tissue permittivity, fat distribution, thoracic geometry or radar cross section on its own. The BMI-dependent relations

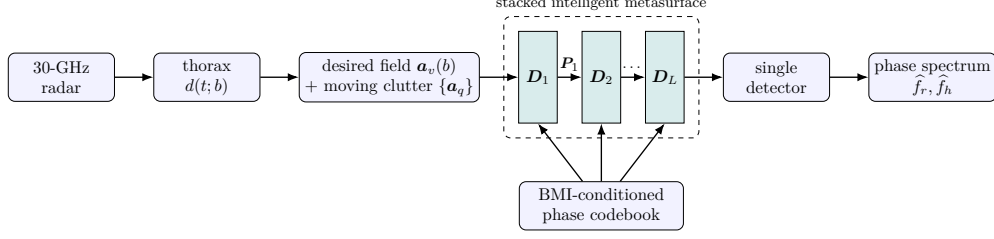


Fig. 1 Proposed morphology-adaptive wave-domain receiver. The phase masks are optimized offline. During sensing, BMI selects a codebook entry and the reflected field is transformed electromagnetically before a single detector.

introduced below are therefore transparent simulation assumptions that can later be replaced by full-wave or measured data.

Against this background, we ask a narrow system question: if morphology changes the spatial signature of the thoracic return, can a morphology-conditioned SIM retain cardiopulmonary micromotion while suppressing spatially distinct moving clutter more effectively than a single universal phase stack? We examine this question with a coupled physiological–electromagnetic digital twin and an exact adjoint gradient for phase-only optimization. The adaptive stack is compared with a single antenna, a 36-channel robust digital array, a one-layer intelligent surface and a universal three-layer SIM under paired Monte Carlo trials. We then vary the receiver SNR, phase resolution, layer loss and BMI-codebook error. The purpose is to establish a reproducible engineering hypothesis and its limits, rather than to claim clinical performance from synthetic data.

2 System and Signal Model

2.1 Architecture

Figure 1 summarizes the receive chain considered here. The illuminating radar is conventional; the change is on reception, where the thoracic return and the nuisance fields encounter the SIM before reaching a single detector. We assume that the controller selects an offline-optimized phase stack from a BMI-indexed codebook. The BMI value is entered during setup or obtained from an existing record. Estimating it from the radar signal is outside the present model.

2.2 Physiological phase modulation

Let b denote the BMI value used for conditioning. We represent anterior chest-wall motion by the sum of respiratory and cardiac components,

$$d(t; b) = A_r(b) \sin(2\pi f_r t + \phi_r(t)) + A_h(b) \sin(2\pi f_h t + \phi_h(t)), \quad (1)$$

where f_r and f_h are the respiration and heartbeat frequencies. Slowly varying terms in $\phi_r(t)$ and $\phi_h(t)$ prevent the simulated waveform from being perfectly periodic. At

wavelength $\lambda = c/f_c$, the corresponding narrowband return is

$$s_v(t; b) = \alpha(b) \exp\left(j \frac{4\pi}{\lambda} d(t; b)\right). \quad (2)$$

where the factor $4\pi/\lambda$ accounts for the two-way path. We deliberately use simple exponential maps for the morphology dependence,

$$\alpha(b) = \exp[-0.035 \max(b - 22, 0)], \quad (3)$$

$$A_r(b) = 4.0 \text{ mm} \exp[-0.012 \max(b - 22, 0)], \quad (4)$$

$$A_h(b) = 0.24 \text{ mm} \exp[-0.020 \max(b - 22, 0)]. \quad (5)$$

These functions are not fitted population relationships. Their role is to create a controlled reduction in return strength and micromotion amplitude across the test range. Keeping them explicit makes it straightforward to replace them when measured or full-wave data become available.

2.3 BMI-conditioned spatial signature

For aperture steering vector $\mathbf{a}(\Omega)$, the desired first-layer field is a coherent mixture of a central and two diffuse components:

$$\begin{aligned} \mathbf{a}_v(b) = \frac{1}{C_b} \bigg[& (1 - \rho_b) \mathbf{a}(\Omega_0(b)) \\ & + 0.55 \rho_b e^{j\psi_b} \mathbf{a}(\Omega_-(b)) \\ & + 0.45 \rho_b e^{-j0.7\psi_b} \mathbf{a}(\Omega_+(b)) \bigg] \odot \mathbf{c}_b, \end{aligned} \quad (6)$$

Here, C_b normalizes the field, ρ_b sets the diffuse fraction and $\mathbf{c}_b$ introduces quadratic phase curvature to mimic a change in effective scattering depth. Independent angular jitter is used during training and testing. This spatial model is important. If the desired return and the clutter were represented by a common scalar phase alone, no static passive linear surface could separate them.

2.4 Multilayer wave-domain transformation

The N elements of layer ℓ implement

$$\mathbf{D}_\ell(\boldsymbol{\theta}_\ell) = \text{diag}(e^{j\theta_{\ell,1}}, \dots, e^{j\theta_{\ell,N}}). \quad (7)$$

For element positions (x_n, y_n) and layer spacing z_ℓ , scalar Green-function coupling is sampled as

$$[\tilde{\mathbf{P}}_\ell]_{mn} = \frac{\exp(jkr_{mn})}{r_{mn}}, \quad (8)$$

$$r_{mn} = \sqrt{(x_m - x_n)^2 + (y_m - y_n)^2 + z_\ell^2}.$$

We do not assume a particular fabricated unit cell at this stage. Instead, the finite coupling matrix is projected onto its closest unitary polar factor $\mathbf{P}_\ell = \mathbf{U}_\ell \mathbf{V}_\ell^H$, where $\tilde{\mathbf{P}}_\ell = \mathbf{U}_\ell \mathbf{\Sigma}_\ell \mathbf{V}_\ell^H$. This retains the Fresnel-like modal mixing while allowing insertion loss to be introduced separately. With field transmission η_ℓ , the layer recursion becomes

$$\begin{aligned} \mathbf{u}_{\ell+1} &= \eta_\ell \mathbf{P}_\ell \mathbf{D}_\ell \mathbf{u}_\ell, \\ \eta_\ell &= 10^{-\xi_\ell/20}, \end{aligned} \quad (9)$$

where ξ_ℓ denotes the loss assigned to layer ℓ in decibels. For output mode $\mathbf{w}$, source q then has the effective gain

$$g_q(\boldsymbol{\Theta}) = \mathbf{w}^H (\eta_L \mathbf{P}_L \mathbf{D}_L) \cdots (\eta_1 \mathbf{P}_1 \mathbf{D}_1) \mathbf{a}_q. \quad (10)$$

2.5 Detector signal

With Q moving nuisance reflectors,

$$\begin{aligned} y(t; b) &= g_v(\boldsymbol{\Theta}, b) s_v(t; b) \\ &+ \sum_{q=1}^Q \sqrt{p_q} g_q(\boldsymbol{\Theta}) c_q(t) + n(t), \end{aligned} \quad (11)$$

where $n(t) \sim \mathcal{CN}(0, \sigma_n^2)$. We include nuisance motion both inside and outside the two vital-sign bands. Removing the temporal mean represents the usual static clutter and I/Q-centre calibration; the time-varying part of each nuisance return is left intact.

3 Adaptive SIM Optimization

3.1 Mean log-SINR objective

The phase masks should strengthen the thoracic field without simply amplifying the moving clutter. For a training set $\mathcal{B}$ of morphology conditions, we define

$$\begin{aligned} S_b &= \alpha^2(b) |g_v(\boldsymbol{\Theta}, b)|^2, \\ I &= \sum_{q=1}^Q p_q |g_q(\boldsymbol{\Theta})|^2 + \sigma_0^2. \end{aligned} \quad (12)$$

and optimize

$$\begin{aligned} \max_{\Theta} \quad & \mathcal{J}(\Theta) = \frac{1}{|\mathcal{B}|} \sum_{b \in \mathcal{B}} \log S_b - \log I, \\ \text{subject to} \quad & \theta_{\ell,n} \in [-\pi, \pi). \end{aligned} \quad (13)$$

For the universal design, $\mathcal{B}$ contains the complete BMI grid and only one stack is trained. For the adaptive design, we optimize a separate Θ_b for every codebook entry and use the nearest entry during sensing. The logarithmic objective was chosen so that conditions with a naturally stronger return do not dominate the universal solution.

3.2 Electromagnetic adjoint gradient

Let $\mathbf{v}_\ell = \mathbf{D}_\ell \mathbf{u}_\ell$. Starting from the detector mode, define

$$\begin{aligned} \mathbf{r}_\ell &= (\eta_\ell \mathbf{P}_\ell)^H \mathbf{q}_{\ell+1}, \\ \mathbf{q}_\ell &= \mathbf{D}_\ell^H \mathbf{r}_\ell. \end{aligned} \quad (14)$$

The exact real phase derivative for one source is

$$\frac{\partial |g|^2}{\partial \theta_{\ell,n}} = 2 \operatorname{Re} \{ g^* j v_{\ell,n} r_{\ell,n}^* \}. \quad (15)$$

The target and clutter terms are combined through the chain rule for Eq. (13). We use Adam [14] for phase ascent and wrap the result into $[-\pi, \pi)$ after every update. Before running the main experiments, we checked the adjoint derivative against a central finite-difference calculation.

3.3 Hardware-constrained masks

For B phase-control bits, continuous coefficients are quantized after training:

$$\begin{aligned} \hat{\theta}_{\ell,n}^{(B)} &= \Delta_B \left\lfloor \frac{\theta_{\ell,n}}{\Delta_B} + \frac{1}{2} \right\rfloor \bmod 2\pi, \\ \Delta_B &= \frac{2\pi}{2^B}. \end{aligned} \quad (16)$$

We apply this rule after continuous-phase training. The resulting experiment therefore measures the penalty of direct hardware quantization; it does not include quantization-aware retraining.

4 Baselines and Rate Estimation

We compare five receivers under the same simulated scenes.

1. **Single antenna:** one central aperture sample and no intelligent spatial processing.
2. **Digital robust MVDR-family array:** a unit-norm complex combiner with one receiver channel per aperture element [15].

3. **One-layer adaptive RIS:** one BMI-conditioned phase mask and one detector.
4. **Universal three-layer SIM:** one mask stack trained over all BMI conditions.
5. **Adaptive three-layer SIM:** a nearest-BMI phase-codebook stack.

For the digital baseline, the nuisance and robust target covariances are

$$\mathbf{R}_i = \sum_q p_q \mathbf{a}_q \mathbf{a}_q^H + \sigma_0^2 \mathbf{I}, \quad (17)$$

$$\mathbf{R}_s(b) = \frac{1}{M} \sum_{m=1}^M \mathbf{a}_{v,m}(b) \mathbf{a}_{v,m}^H(b). \quad (18)$$

The dominant generalized eigenvector of $\mathbf{R}_s \mathbf{w} = \lambda \mathbf{R}_i \mathbf{w}$ provides the initial combiner. We then refine it on the complex unit sphere using the same mean log-SINR objective as the SIM. This is intentionally a demanding reference: it has access to all 36 complex spatial channels, whereas each surface-based receiver terminates in one detector.

Rate estimation is kept deliberately simple so that differences are driven mainly by the spatial front end. We unwrap and linearly detrend the received phase, apply a Hann window, and calculate an eightfold zero-padded FFT. Quadratic interpolation of the log-spectrum gives the largest peak in 0.10–0.60 Hz for respiration and 0.80–2.20 Hz for heartbeat. Across K trials, the rate error is

$$\text{RMSE}_x = 60 \sqrt{\frac{1}{K} \sum_{k=1}^K (\hat{f}_{x,k} - f_{x,k})^2}, \quad (19)$$

$x \in \{r, h\}.$

5 Methods

The reference configuration is collected in Table 1. We fixed the random seed at 2026 and used paired trials: every receiver sees the same physiological frequencies, spatial jitter, nuisance phases and normalized noise realization. This removes scene-to-scene variation from comparisons between receivers. The reference SNR γ is defined using the desired power at one aperture element,

$$\sigma_n^2 = \frac{\alpha^2(b)}{N} 10^{-\gamma/10}. \quad (20)$$

so aperture gain can emerge from coherent collection without multiplying the noise power by the number of metasurface elements.

For the main five-receiver comparison, propagation between the phase masks is lossless. This isolates the spatial degrees of freedom of each architecture. Loss is introduced separately in the depth study at 0.5 dB per layer. We report spatial performance as

$$\Gamma_b = 10 \log_{10} \left(\frac{\alpha^2(b) |g_v|^2}{\sum_q p_q |g_q|^2 + \sigma_0^2} \right). \quad (21)$$

Table 1 Reference simulation parameters

Parameter	Value
Carrier frequency	30 GHz
Wavelength	9.99 mm
SIM aperture	6×6 elements ($N = 36$)
Element pitch	5 mm
Inter-layer spacing	12 mm
Sampling rate / duration	20 Hz / 30 s
BMI codebook	$\{18, 22, 26, 30, 34, 38\}$ kg/m ²
Respiration frequency	$\mathcal{U}(0.18, 0.42)$ Hz
Heart frequency	$\mathcal{U}(0.95, 1.65)$ Hz
Nuisance fields	8 moving spatial sources
Training variants	4/BMI universal; 6/BMI adaptive
Optimizer	Adam, 180 iterations, step 0.045
Main evaluation	10 trials per BMI at 8 dB
Sensitivity evaluation	12 trials per condition
SNR sweep	-10, -5, 0, 5, 10, 15 dB
Phase resolution	1, 2, 3, 4, 6 bits; continuous
Depth / assumed loss	1-5 layers / 0.5 dB per layer
BMI-selection bias	-8, -4, 0, 4, 8 kg/m ²

The implementation is also checked with five automated tests covering field normalization, clean-signal rate recovery, phase quantization, codebook remapping and the adjoint gradient.

6 Results

6.1 Reference comparison

Table 2 gives the mean over all six BMI conditions at the 8-dB reference SNR. In this simulation, the adaptive SIM reaches 9.29 dB spatial SINR, which is 3.57 dB above the adaptive one-layer surface, 0.91 dB above the universal three-layer stack and 0.03 dB below the 36-channel digital array. These are descriptive comparisons for the specified model and fixed random seed, rather than statistical evidence of general superiority. The small difference from the digital array is a property of the ideal modal model: the layers and the output port together can realize an effective complex spatial combiner. We do not expect this equality to survive unchanged once measured loss, coupling, bandwidth and calibration error are included.

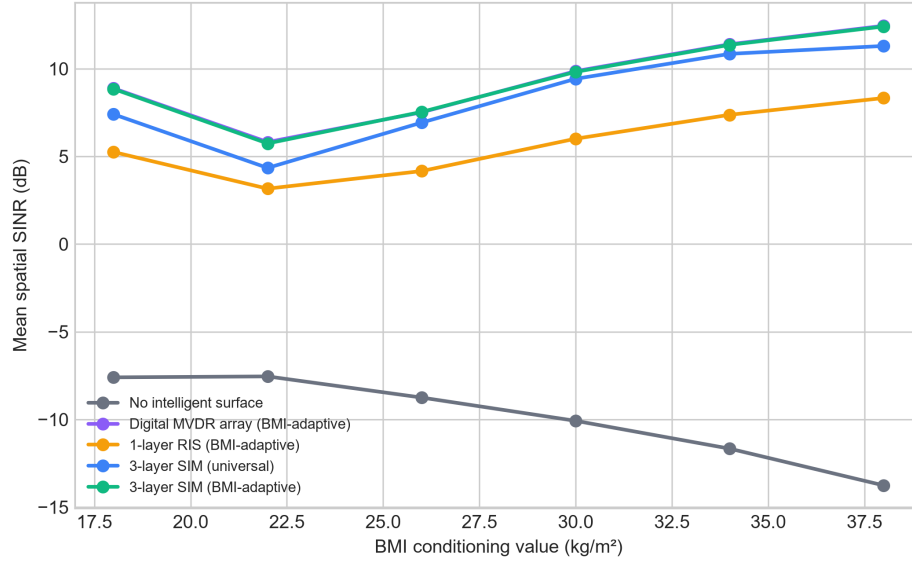
Spatial SINR does not tell the whole story. Cardiac RMSE depends on where the residual clutter falls in frequency, not only on its total power. In this seeded run, the digital and universal receivers each fail at one BMI condition, while the adaptive SIM remains stable. We regard this as a feature of the particular ensemble rather than evidence of a general advantage; substantially more trials would be needed to support the latter claim.

Figure 2 separates the results by conditioning value. As the assumed target attenuation increases, the single-antenna baseline deteriorates. The adaptive SIM follows the digital array closely and remains above the universal stack, with the largest separation at the lower edge of the BMI grid and around BMI 22.

Table 2 Reference results at 8-dB single-antenna SNR (simulation only)

Method	SINR (dB)	Respiration RMSE	Heart RMSE
No intelligent surface	−9.90	13.29	22.31
Digital robust MVDR array (BMI-adaptive)	9.32	0.05	1.09
One-layer RIS (BMI-adaptive)	5.72	0.05	0.16
Three-layer SIM (universal)	8.38	0.05	1.16
Three-layer SIM (BMI-adaptive)	9.29	0.05	0.08

RMSE values are in breaths/min for respiration and beats/min for heart rate.

**Fig. 2** Mean spatial SINR versus BMI conditioning value for the five receiver architectures. The mapping from BMI to field response is parametric and unvalidated.

6.2 SNR robustness

The SNR sweep in Fig. 3 shows a marked transition between 0 and 5 dB under the tested conditions. At 0 dB, the adaptive SIM recovers respiration reasonably well, with an RMSE of 1.62 breaths/min, but its heart-rate RMSE is still 10.26 beats/min. At 5 dB these errors fall to 0.05 breaths/min and 0.12 beats/min. Below −5 dB, every receiver frequently locks onto the wrong spectral peak. Ranking the methods by their already large errors in that region would therefore be misleading.

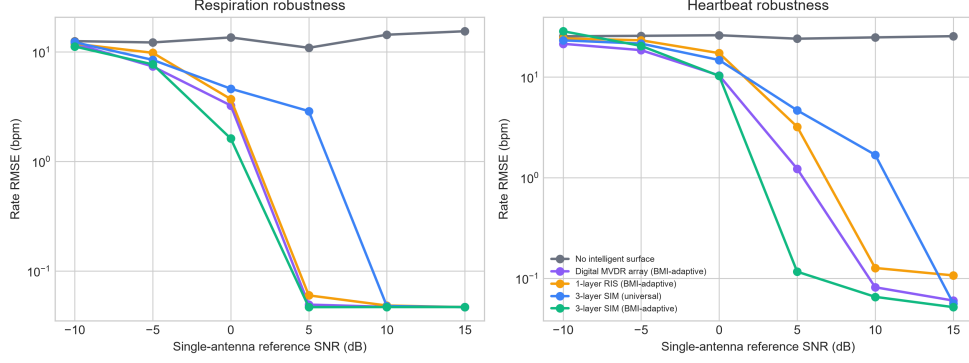


Fig. 3 Rate-estimation robustness versus single-antenna reference SNR. The logarithmic ordinate exposes the estimator transition between 0 and 5 dB.

6.3 Finite phase resolution

Directly rounding the optimized phases is costly below four bits (Fig. 4). With six-bit control, the mean SINR is 9.00 dB and the heart-rate RMSE is 0.08 beats/min, close to 9.25 dB and 0.08 beats/min for continuous phase. Four-bit control falls to 5.70 dB and 1.91 beats/min. The one- and two-bit points are not monotonic because coarse, independent rounding can either preserve or remove a particular interference null by chance. Neither resolution is reliable in this experiment. [The result applies to post-training phase rounding and does not establish a general hardware-resolution requirement.](#)

6.4 Depth–loss tradeoff

Once each layer is assigned 0.5 dB insertion loss, a deeper stack is no longer always better. Two layers give the best point in the tested set, at 8.29 dB SINR and 0.08 beats/min heart-rate RMSE (Fig. 5). A third layer adds controllable phases but also another 0.5 dB loss, reducing the SINR to 8.08 dB. The location of this optimum is tied to the assumed loss and should not be treated as a universal two-layer rule.

6.5 Conditioning mismatch

If b is the simulated morphology and $\hat{b} = b + \delta_b$ is supplied to the controller, the selected code is

$$c(\hat{b}) = \arg \min_{\beta \in \mathcal{B}_c} |\hat{b} - \beta|, \quad (22)$$

$$\Theta = \Theta_{c(\hat{b})}.$$

In Fig. 6, the simulated subject is held fixed and only δ_b is changed. Selecting a code with a bias of -8 kg/m^2 costs 2.41 dB relative to the unbiased case. A $+8 \text{ kg/m}^2$ bias still gives 7.29 dB mean SINR, yet the heart-rate RMSE rises to 5.89 beats/min. This is another example of residual clutter structure mattering more than total interference power. The observed asymmetry comes from the finite codebook and the chosen morphology map; it is not a claim about physiological asymmetry.

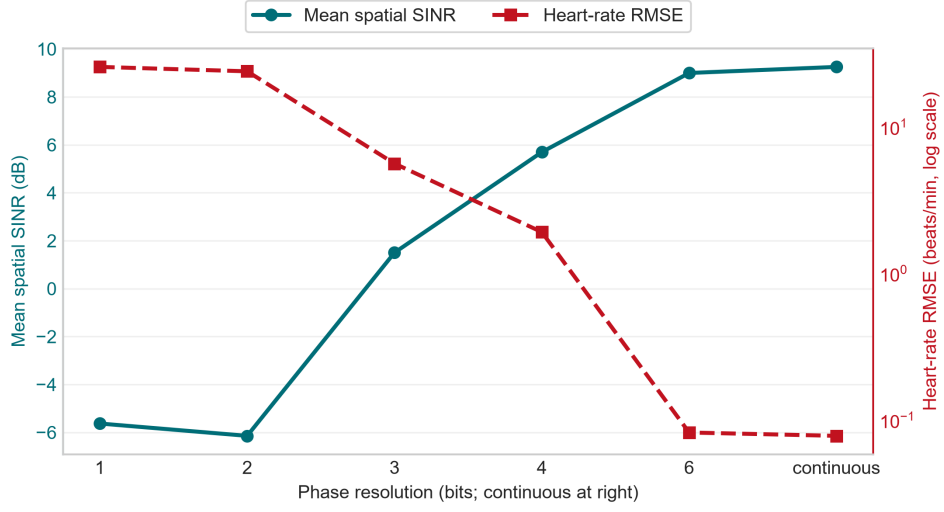


Fig. 4 Adaptive three-layer SIM performance after post-training phase quantization. Mean spatial SINR uses the left axis and heart-rate RMSE uses the logarithmic right axis. Quantization-aware retraining is not used.

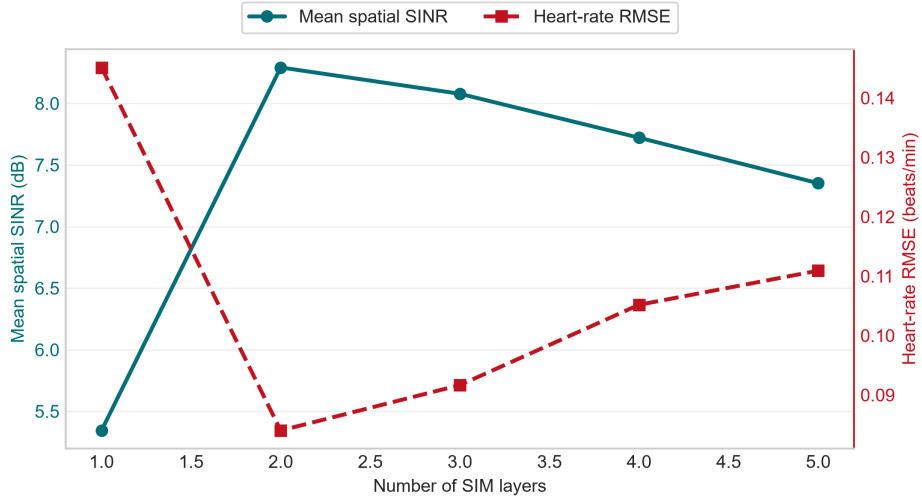


Fig. 5 Layer-depth sensitivity with 0.5 dB insertion loss per layer. Mean spatial SINR uses the left axis and heart-rate RMSE uses the right axis.

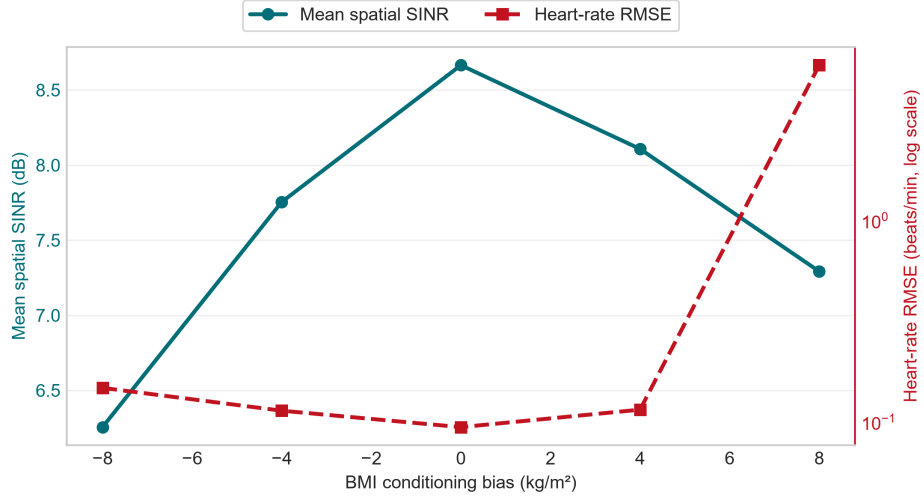


Fig. 6 Sensitivity to biased BMI codebook selection at 8-dB reference SNR. Mean spatial SINR uses the left axis and heart-rate RMSE uses the logarithmic right axis.

7 Discussion

Three observations are particularly relevant to a practical design. First, [under the single-output narrowband model, the simulated passive stack produced a spatial SINR close to that of the fully digital array while retaining one detector.](#) The useful comparison is therefore between RF-chain complexity and electromagnetic implementation loss, not between SIM and unconstrained digital processing in the abstract. Second, morphology conditioning has value only when changes in the desired field affect its separation from nearby clutter modes. If those modes are already well separated, a universal stack is the simpler choice. Third, hardware detail can outweigh nominal network depth. Here, six-bit control behaves much like continuous phase, whereas [the assumed accumulated loss makes the two-layer design the best of the tested depths.](#)

The word “neural” also needs care. Equations (9) and (10) describe a trainable multilayer operator, and its phases are optimized by backpropagating an adjoint field. Its selectivity nevertheless comes from coherent interference in a linear passive structure. The nonlinear operations occur after the detector, in phase recovery and rate estimation.

7.1 Limitations and required validation

The largest uncertainty is the absence of measured, morphology-conditioned electromagnetic data. BMI cannot resolve fat distribution, waist circumference, posture, thoracic geometry, clothing or tissue permittivity. Equations (3)–(6) should therefore be regarded as placeholders for a relationship that still has to be measured. They would

need to be calibrated with anatomically representative phantoms or synchronized human measurements before supporting any population-level conclusion.

The propagation model is also intentionally idealized. It omits mutual coupling, finite-aperture edge effects, frequency dispersion, angle-dependent element response, phase–amplitude coupling, fabrication tolerance, controller latency and thermal drift. The scene contains one subject and simplified moving reflectors, static-clutter calibration is assumed to be effective, and the rate estimator is deliberately basic. The Monte Carlo ensemble demonstrates reproducibility of the code and trends; it is not large enough to establish clinical accuracy or limits of agreement. [The comparisons are descriptive and use a single fixed random seed; the study does not report confidence intervals or formal statistical tests and therefore does not establish a general ranking of the receiver architectures.](#)

The immediate electromagnetic follow-up is well defined. A programmable unit cell can be simulated over frequency, phase state, incidence angle and polarization to obtain its complex reflection and transmission coefficients. Those data would replace the ideal diagonal mask and the assumed scalar loss with measured or full-wave frequency-dependent operators. A finite multilayer model could then quantify mutual coupling, edge diffraction and the loss–depth tradeoff before a surface is fabricated. After that calibration, the frozen designs should be tested first on phantoms, then on public synchronized radar/ECG/respiration datasets [12, 13], and finally in an appropriately approved human-subject study.

8 Reproducibility

GitHub repository: <https://github.com/shujaansari/SIM-VITAL-LINK-6G>

9 Conclusion

We have examined whether a stacked intelligent metasurface can adapt its receive-side field transformation to a simple morphology condition for contactless respiration and heartbeat sensing. [Within the present digital twin and its idealized assumptions, the adaptive three-layer SIM produced spatial SINR within 0.03 dB of the 36-channel robust digital array and higher mean spatial SINR than the one-layer surface and universal stack in the tested ensemble. This descriptive result should not be read as evidence of general hardware or clinical superiority.](#) The sensitivity studies expose the engineering constraints behind that result: rate recovery becomes reliable only above the observed SNR transition, coarse phase control is damaging, and the assumed layer loss can make a shallower stack preferable. These findings motivate full-wave and phantom validation, but they do not replace either. The next step is to substitute electromagnetic S-parameters and measured morphology data for the ideal operators used here.

Declarations

Author contributions. All authors contributed to the conception and development of the study, preparation of the initial manuscript, evaluation of the results, critical

revision of the article, and approval of the submitted version. All authors accept responsibility for the integrity and accuracy of the work.

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Use of generative artificial intelligence. The scientific argument, modelling choices, simulation results and interpretation remain the responsibility of the authors. During preparation, the authors used OpenAI ChatGPT/Codex to assist with software and workflow setup, presentation of mathematical material, organization of some passages and language review. No AI system was treated as an author or as a source of scientific evidence. The authors checked the code, equations, citations, numerical results and final text and accept responsibility for the accuracy and integrity of the manuscript.

Data availability. This study used no human-participant or clinical data. All data analysed in the manuscript are synthetic outputs generated by the simulation model described in the Methods. The generated data underlying the figures and tables are available in the **results** and **sensitivity_results** directories of the public GitHub repository at <https://github.com/shujaansari/SIM-VITAL-LINK-6G>.

Code availability. The source code, run scripts, fixed random seeds and environment requirements are available at <https://github.com/shujaansari/SIM-VITAL-LINK-6G>.

Ethics approval and consent to participate. Not applicable. This theoretical and simulation-based study did not involve human participants, human tissue, animals, clinical records, or identifiable patient data.

Competing interests. The authors declare no competing interests.

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